

SHIFAN JIA

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Data Scientist with hands-on experience in production **AI systems, machine learning, LLM evaluation, AI agents, deep learning, and statistical modeling**. Passionate about developing reliable AI solutions for real-world healthcare and business applications.

Education

Simon Fraser University

Ph.D. in Statistics

09/2022 - 05/2026

Vancouver, Canada

George Washington University

MSc in Statistics

09/2019 - 02/2021

Washington D.C, USA

Industry Experience

Fraser Health, Data Scientist

03/2026 - now

Vancouver, BC

- Designed and deployed production AI applications using Python and Azure Machine Learning to improve healthcare service routing and AI safety.
- Developed AI-powered decision support pipelines combining deterministic models and LLM reasoning to improve routing quality and operational efficiency.
- Designed experimentation frameworks using A/B testing, statistical analysis, and offline evaluation to measure LLM product performance and support data-driven decisions.
- Partnered with product managers and stakeholders to define KPIs, monitor product performance, and deliver executive-ready recommendations.

Scotiabank, Data Scientist Intern

09/2024 - 12/2024

Toronto, ON

- Analyzed Canadian Agriculture write-off data (2006–2023) using Python, SQL, Pandas, and NumPy to evaluate portfolio performance and risk trends.
- Designed and implemented a Transformer-based model by integrating pre-trained BERT with custom neural network layers to enhance Scotiabank's risk rating system, using PyTorch.
- Applied the model to analyze override comments and predict rating adjustments, achieving 70% consistency with human evaluations and reducing the need for manual review.

Technical Skills

- **Programming:** Python, SQL, R
- **AI & Machine Learning:** LLM Evaluation, AI Agents, NLP, Deep Learning, Supervised & Unsupervised Learning, Statistical Modeling, A/B testing
- **Frameworks & Libraries:** DSPy, PyTorch, Scikit-learn, Pandas, NumPy, MLflow
- **Cloud & MLOps:** Azure Machine Learning, Azure AI Foundry, Azure AI Content Safety, Git, GitHub Actions, AWS

Selected Research Projects

Transformer-based Credit Risk Assessment

- Developed a transformer-based NLP framework using BERT to analyze credit override documents for risk rating prediction.
- Combined contextual text embeddings with structured financial features and applied Lasso-based feature selection to improve predictive performance.
- Achieved over **90% prediction accuracy**, supporting more consistent and reliable credit risk assessments.

Multimodal Deep Learning for Competing Risks Survival Prediction

- Designed a multimodal deep learning framework integrating EHR data, ICU clinical notes, and longitudinal functional data for competing-risks survival prediction.
- Leveraged pre-trained language model embeddings, self-attention mechanisms, and B-spline basis functions to model continuous-time survival probabilities.
- Developed one of the first deep survival models capable of continuous probability prediction beyond traditional discrete-time approaches.

Maternal–Infant Weight Modeling using Functional Regression

- Developed a functional regression model to predict newborn weight from maternal BMI trajectories collected over irregular time domains.
- Designed kernel-weighted estimation with B-spline basis expansions and translated the model into an interactive R Shiny application for clinical decision support.
- Provided interpretable insights to support pregnancy weight management and maternal healthcare.

Research Paper

- **Shifan Jia**, Haolun Shi and Tianyu Guan, “Function-on-Function Regression Models with Nonlinear Dynamic Effect and Linear Concurrent Effect (2025)”, **Statistics and Computing**.
- **Shifan Jia**, Ruifen Zhao, Li Zhou, Haolun Shi and Tianyu Guan. “Variable-Domain Weighted Scalar-on-Function Regression for Maternal-Infant Weight Modeling (2025)” **under review by Biometrics**.
- **Shifan Jia**, Haolun Shi, Haixu Wang and Tianyu Guan. ‘M2M-FSurv: Multimodal Deep Functional Survival Analysis with Multiple Competing Risks’, submitted to ICML.
- Yuming Sun, Lufeng Feng, Baomin Xu, **Shifan Jia**, Li Duan, Wei Ni, and Ziyu Jia. “Alzheimer’s Detection Based on EEG Source Imaging and Frequency-Joint Attention Fusion Network (2025)”. **Journal of Neural Engineering**.
- Ruoyu Wang, Lufeng Feng, **Shifan Jia**, Li Duan and Baomin Xu. “MorphSys: Branch-Aware Contrastive Learning for Neuron Morphology Graphs (2025)”, **under review by Journal of Neural Engineering**.
- Lufeng Feng, **Shifan Jia**, Li Duan and Baomin Xu. “Supervised Contrastive Learning-Based Adaptive Multi-Scale Time-Frequency Network for Motor Imagery Decoding (2025)”, **Journal of Biomedical Signal Processing and Control**
- Lufeng Feng, Yueru Liu, Baomin Xu, **Shifan Jia**, Wei Ni, Li Duan. “A New Dual-Attention Multi-Node Fusion Network for EEG-fNIRS Motor Imagery Classification (2025)”, **under review by Journal of Biomedical and Health Informatics**.
- Sidi Tao, Lufeng Feng, **Shifan Jia**, Shuangyuan Yu, Baomin Xu. “BiGSTF-Net: Inter-modal Mutual Guidance and Intra-modal Spatio-temporal Fusion for EEG-fNIRS Cognitive Tasks Classification (2025). **under review by Journal of Biomedical Signal Processing and Control**